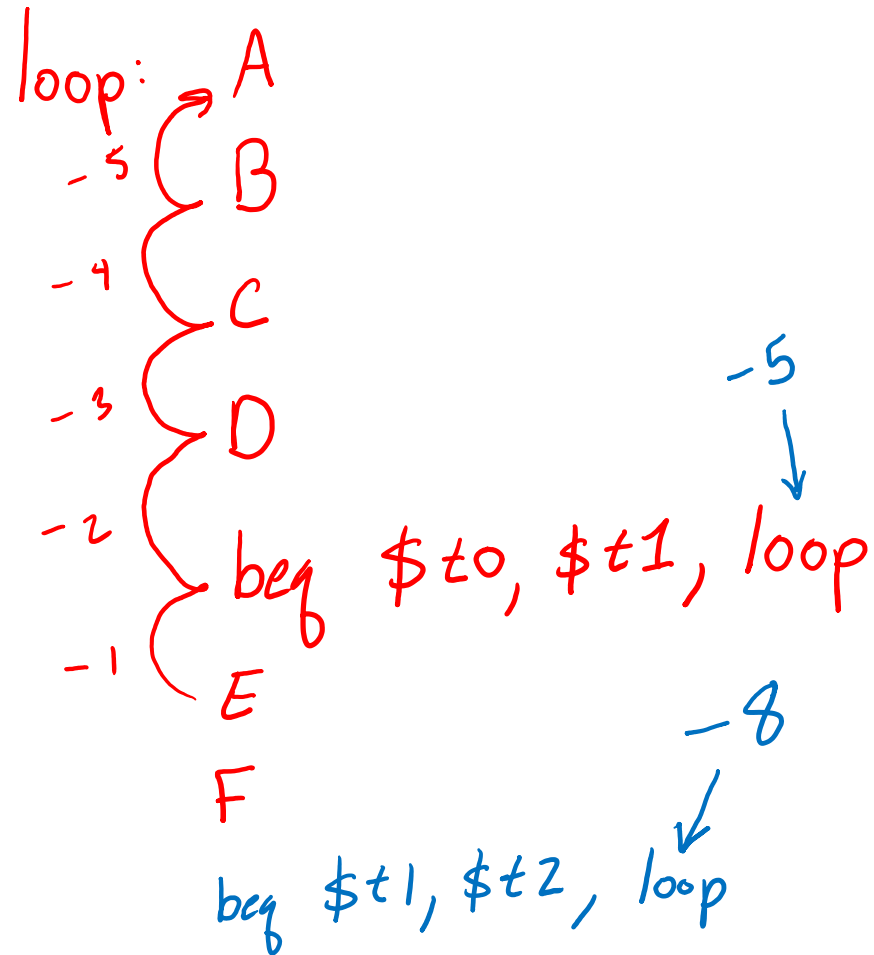


Branches

- Lab today in 14-303
- must work with a partner on lab
- check code into github

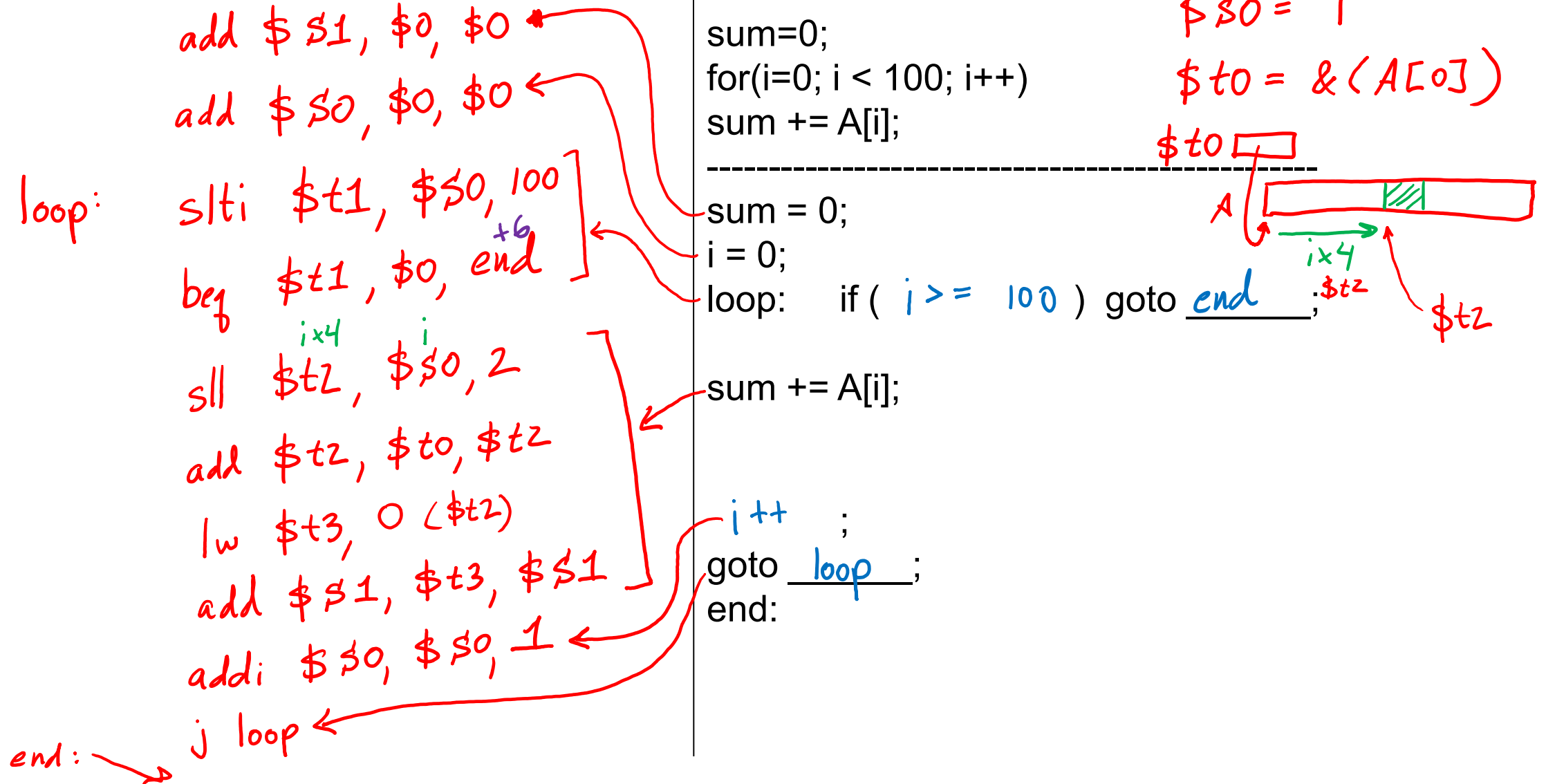
Operation	rs	rt	rd	shamt	funct	# comment
beq	3	2	-5			# if (\$2 == \$3) goto loop
bne	3	2				# if (\$2 != \$3) goto loop
jr		3			8	# goto \$3
j						# goto loop

Computing Branch Targets



1. start at inst. after branch
2. count to the branch target (if upward, it is negative)

MIPS Example 4



MIPS Instructions

- destination first
- immediates are limited to 16-bit
- insts. are 32-bits in size

- Lab 1 (due 4/11 11:59p)

spim

/home/jseng/bin/spim -f file.asm

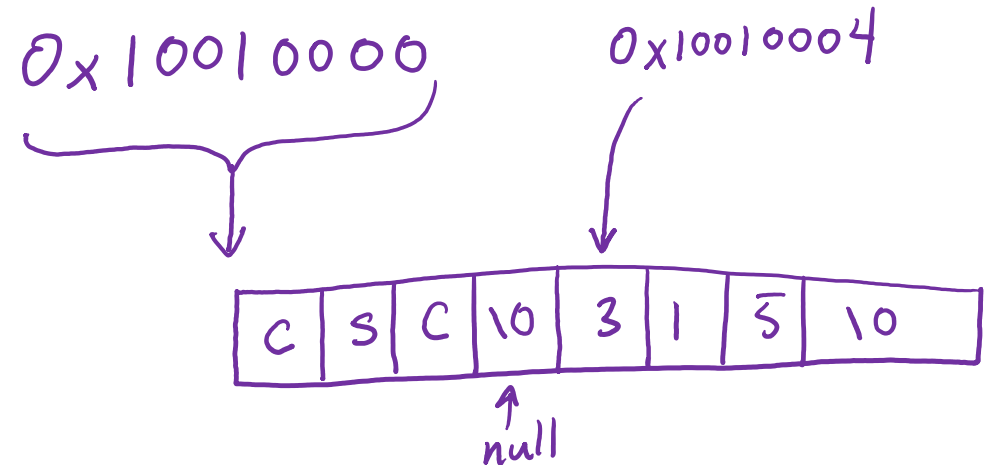
- work with a partner
- store code in private github
- progress check on Wed.

- syscall is used for I/O

\$V0 ← 1 = print an integer in \$a0
\$V0 ← 4 = print a string (start address of string is in \$a0)
\$V0 ← 5 = read in an integer

- the start of all strings is at 0x10010000

- do not use 'la' inst.



file.asm

